

Prime Multiples

Time limit: 1.00 s

Memory limit: 512 MB

You are given k distinct prime numbers $a_1, a_2, \dots, a_k$ and an integer n .

Your task is to calculate how many of the first n positive integers are divisible by at least one of the given prime numbers.

Input

The first input line has two integers n and k .

The second line has k prime numbers $a_1, a_2, \dots, a_k$.

Output

Print one integer: the number integers within the interval $1, 2, \dots, n$ that are divisible by at least one of the prime numbers.

Constraints

- $1 \leq n \leq 10^{18}$
- $1 \leq k \leq 20$
- $2 \leq a_i \leq n$

Example

Input	Output
20 2 2 5	12

Explanation: the 12 numbers are 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18, 20.